Lecture 8

Highest weights of representations of compact groups

Objectives. *The Lie Algebras course classified finite-dimensional irreducible modules of semisimple algebras by dominant integral weights. This lecture adds the action of the centre of the Lie algebra and decides which of these modules are representations of a given compact connected Lie group K. These are exactly the modules whose highest weight is a character of a maximal torus.*

Lecture 7 decided, for a given highest-weight module, whether it descends. It did not show that every irreducible representation of K is obtained in this way, or that one is obtained only once. This lecture proves both.

By the end of this lecture you will be able to:

- Prove that highest weights give a bijection $\mathrm{Irr}(K) \to X^*(T)^+$, assuming cited structure results.
- Decide descent to K_{sc}/F by evaluating a highest weight λ on the elements $H \in \mathfrak{t}$ with $\exp H \in F$.
- Treat a central torus by an integral character and a finite compatibility condition.
- Explain why some modules of a reductive Lie algebra come from no compact group.

8.1 Torus characters and dominance

Throughout, K is a compact connected Lie group with Lie algebra $\mathfrak{k}$. Let T be a maximal torus with Lie algebra $\mathfrak{t}$ (Lecture 6). Put $\Lambda = \ker(\exp\colon \mathfrak{t} \to T)$. A *character* of T is a continuous homomorphism $\chi\colon T \to \mathrm{U}(1)$. Its derivative has the form $d\chi = i\ell$ for some $\ell \in \mathfrak{t}^*$, and

$$\chi(\exp H) = e^{i\ell(H)} \quad (H \in \mathfrak{t}), \qquad \ell(\Lambda) \subset 2\pi\mathbb{Z}. \tag{8.1}$$

Conversely, $\exp\colon \mathfrak{t} \to T$ is a surjective homomorphism with kernel Λ. So each ℓ with $\ell(\Lambda) \subset 2\pi\mathbb{Z}$ defines a character. Let $X^*(T)$ be the lattice of these ℓ.

Let Π be a representation of K on V. By Lecture 5, V has a K-invariant inner product. So the operators $\Pi(t)$ with $t \in T$ are commuting unitary operators. Hence these operators have a common orthonormal eigenbasis. For a common eigenvector v, write $\Pi(t)v = c(t)v$. Then c

is a character of T, so $c(\exp H) = e^{i\ell(H)}$ for a unique $\ell \in X^*(T)$. Hence V is the direct sum of the weight spaces

$$V_\ell = \{v \in V : \ \Pi(\exp H)v = e^{i\ell(H)}v \ \text{ for all } H \in \mathfrak{t}\}, \qquad \ell \in X^*(T).$$

Differentiating $\Pi(\exp tH)v = e^{it\ell(H)}v$ at $t = 0$ gives $d\Pi(H)v = i\ell(H)v$ for $v \in V_\ell$. The complex-linear extension of $i\ell$ to $\mathfrak{t}_\mathbb{C}$ is a weight in the sense of the Lie Algebras course. We identify ℓ with this weight.

Roots are the nonzero weights of Ad on $\mathfrak{k}_\mathbb{C}$. Thus $[H, X] = i\alpha(H)X$ for $H \in \mathfrak{t}$ and X in the root space of α. In particular, roots lie in $X^*(T)$.

Fix positive roots R^+ as in [I]. Put

$$\mathbb{N}R^+ = \Big\{ \sum_{\alpha \in R^+} n_\alpha \alpha : \ n_\alpha \in \mathbb{Z}_{\geq 0} \Big\}, \qquad \ell' \preceq \ell \iff \ell - \ell' \in \mathbb{N}R^+.$$

There is $H_+ \in \mathfrak{t}$ with $\alpha(H_+) > 0$ for all $\alpha \in R^+$. So every nonzero element of $\mathbb{N}R^+$ is positive at H_+. Hence $\ell' \preceq \ell$ and $\ell \preceq \ell'$ force $\ell = \ell'$. A *highest weight* of Π is a weight ℓ of Π with $\ell' \preceq \ell$ for every weight ℓ' of Π. It is unique if it exists.

Write $\mathfrak{k} = \mathfrak{z} \oplus \mathfrak{k}_s$, with $\mathfrak{z}$ the centre and $\mathfrak{k}_s = [\mathfrak{k}, \mathfrak{k}]$ semisimple. Then $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ with $\mathfrak{t}_s = \mathfrak{t} \cap \mathfrak{k}_s$ (Lecture 6). Roots vanish on $\mathfrak{z}$, since $\operatorname{ad} Z = 0$ for $Z \in \mathfrak{z}$.

The highest-weight theorem of the Lie Algebras course is stated for a Cartan subalgebra of $\mathfrak{k}_{s,\mathbb{C}}$. We now check that $\mathfrak{t}_{s,\mathbb{C}}$ is one, so that the theorem applies here.

The algebra $\mathfrak{t}_s$ is maximal abelian in $\mathfrak{k}_s$. Indeed, let $\mathfrak{a} \supset \mathfrak{t}_s$ be an abelian subalgebra of $\mathfrak{k}_s$. Since $\mathfrak{z}$ is central, $\mathfrak{z} \oplus \mathfrak{a}$ is abelian and contains $\mathfrak{t}$. The Lie algebra of a maximal torus is maximal abelian (Lecture 6), so $\mathfrak{z} \oplus \mathfrak{a} = \mathfrak{t}$. Hence $\mathfrak{a} \subset \mathfrak{t} \cap \mathfrak{k}_s = \mathfrak{t}_s$.

The adjoint operators of elements of $\mathfrak{t}_s$ commute and are skew-adjoint, hence simultaneously diagonalisable over $\mathbb{C}$. So the complexification $\mathfrak{t}_{s,\mathbb{C}}$ is toral. It is also maximal toral. If $X \in \mathfrak{k}_s$ commutes with $\mathfrak{t}_s$, then $\mathfrak{t}_s + \mathbb{R}X$ is abelian, so $X \in \mathfrak{t}_s$. Now let $X, Y \in \mathfrak{k}_s$ with $[H, X + iY] = 0$ for all $H \in \mathfrak{t}_s$. The brackets $[H, X]$ and $[H, Y]$ lie in $\mathfrak{k}_s$, so both vanish, and $X, Y \in \mathfrak{t}_s$. Hence $\mathfrak{t}_{s,\mathbb{C}}$ is its own centraliser in $\mathfrak{k}_{s,\mathbb{C}}$. A toral subalgebra $\mathfrak{h} \supset \mathfrak{t}_{s,\mathbb{C}}$ is abelian [I], so $\mathfrak{h}$ lies in this centraliser and $\mathfrak{h} = \mathfrak{t}_{s,\mathbb{C}}$. Thus $\mathfrak{t}_{s,\mathbb{C}}$ is a maximal toral, or Cartan, subalgebra of $\mathfrak{k}_{s,\mathbb{C}}$. This is the Cartan subalgebra used in [I].

For a root α, [I] supplies a coroot $h_\alpha \in \mathfrak{t}_{s,\mathbb{C}}$. The corresponding root in the sense of the Lie Algebras course is the complex-linear extension of $i\alpha$. This extension takes the value 2 at h_α. By Lecture 7, the real coroot is $H_\alpha = ih_\alpha \in \mathfrak{t}_s$. Since $h_\alpha = -iH_\alpha$, the value of the complex-linear extension of $i\alpha$ at h_α equals $\alpha(H_\alpha)$. So

$$\alpha(H_\alpha) = 2, \qquad H_\alpha \in \mathfrak{t}_s.$$

In the same way, the complex-linear extension of $i\ell$ takes the value $\ell(H_\alpha)$ at h_α. So we put

$$\langle \ell, \alpha^\vee \rangle = \ell(H_\alpha), \qquad P = \{\lambda \in \mathfrak{t}_s^* : \langle \lambda, \alpha^\vee \rangle \in \mathbb{Z} \ \forall \alpha\},$$

$$X^*(T)^+ = \{\ell \in X^*(T) : \langle \ell, \alpha^\vee \rangle \geq 0 \ \forall \alpha \in R^+\}.$$

By Lecture 7, $2\pi H_\alpha \in \Lambda$. Hence every $\ell \in X^*(T)$ satisfies

$$\langle \ell, \alpha^\vee \rangle = \tfrac{1}{2\pi}\, \ell(2\pi H_\alpha) \in \mathbb{Z},$$

so $\ell|_{\mathfrak{t}_s} \in P$. Let Q be the root lattice, the lattice generated by the roots. Roots lie in $X^*(T)$, so for semisimple K we obtain $Q \subset X^*(T) \subset P$. If K is also simply connected, then Λ is generated as a lattice by the $2\pi H_\alpha$, and $X^*(T) = P$ (Lecture 7).

For SU(2), put $h = \operatorname{diag}(1, -1)$, so that ih is the matrix H_0 of Lecture 7. Then $\mathfrak{t} = \mathbb{R}\, ih$, with

$$\exp(\theta\, ih) = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}, \qquad \Lambda = 2\pi\mathbb{Z}\, ih.$$

Since $[\theta\, ih, E_{12}] = 2i\theta E_{12}$, we get $\alpha(ih) = 2$ and $h_\alpha = h$. So $X^*(T) = P = \{\ell : \ell(ih) \in \mathbb{Z}\}$, and dominance says that $m = \ell(ih) \geq 0$. The classification theorem below says that this m labels V_m. For SO(3) only the even m remain.

8.2 The classification theorem

We prove the theorem by passing to a finite cover of K. First let V be irreducible for K. By Schur's lemma, the centre $\mathfrak{z}$ acts by a scalar $i\mu$. So V stays irreducible under $\mathfrak{t}_s$, and the highest-weight theorem of the Lie Algebras course gives a highest weight λ. Thus $\ell = \mu + \lambda$ is defined. A representation of a connected group is determined by its differential, so ℓ determines V.

For existence, start from ℓ. Integrate the module $V(\lambda)$ to the simply connected group K_s with Lie algebra $\mathfrak{t}_s$. Multiply by the character χ_μ of the central torus $Z_0 = \exp \mathfrak{z}$ given by $\chi_\mu(\exp Z) = e^{i\mu(Z)}$. This gives a representation of the finite cover $Z_0 \times K_s$ of K. It descends if and only if the finite kernel F acts trivially. By the lifting lemma, Lemma 8.2, each element of F is the exponential of a point $H \in \Lambda$. It acts on a highest weight vector by $e^{i\ell(H)} = 1$, since $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. By Schur's lemma it then acts trivially.

We use the following structure inputs.

(S1) (Torus theorem) In a compact connected Lie group, every element lies in a maximal torus, and all maximal tori are conjugate (Lecture 6). The image of exp generates the group (Lecture 2). A compact connected abelian Lie group is the image of its Lie algebra under exp, and it is isomorphic to $\mathrm{U}(1)^n$ (Lecture 7). The Lie algebra of a maximal torus is a maximal abelian subalgebra (Lecture 6).

(S2) (Lecture 7; H Chapter 13; covering theorem) Let $Z_0 = \exp \mathfrak{z}$, a torus. Let K_s be the simply connected Lie group with Lie algebra $\mathfrak{t}_s$. Then K_s is compact, by Weyl's theorem on the fundamental group (Lecture 7). The inclusion $\mathfrak{t}_s \subset \mathfrak{t}$ integrates to $\iota\colon K_s \to K$. The map

$$p\colon \widetilde{K} = Z_0 \times K_s \to K, \qquad p(z, s) = z\, \iota(s),$$

is a surjective homomorphism with finite central kernel F, and $dp = \mathrm{id}_{\mathfrak{t}}$. Hence p is a covering homomorphism by Lecture 4. In the notation of Lecture 7, $Z_0 = T_0$, $K_s = K_{sc}$ and $\iota = \varphi$. The group $\widetilde{K}$ is a finite cover of K. It is not the universal cover when $\mathfrak{z} \neq 0$, because Z_0 is a torus. For this reason a character of Z_0 must satisfy an integrality condition.

(S3) (Lecture 4; integration and descent) Finite-dimensional representations of $\mathfrak{t}_s$ integrate to K_s. Let $\kappa\colon G_1 \to G_2$ be a covering homomorphism with kernel N. A representation $\widetilde{\Pi}$ of G_1 equals $\Pi \circ \kappa$ for a representation Π of G_2 if and only if $\widetilde{\Pi}(N) = \{I\}$.

(S4) ([I], highest-weight theorem) Finite-dimensional $\mathfrak{k}_{s,\mathbb{C}}$-modules are sums of weight spaces with weights in P. An irreducible such module has a unique highest weight λ. Its weight space is a line, every weight lies in $\lambda - \mathbb{N}R^+$, and $\langle \lambda, \alpha^\vee \rangle \geq 0$ for $\alpha \in R^+$. Each dominant $\lambda \in P$ is the highest weight of exactly one irreducible module $V(\lambda)$, up to isomorphism.

Lemma 8.1. *Let G be a compact connected Lie group with a maximal torus S and $\mathfrak{s} = \operatorname{Lie} S$. Every central element f of G equals $\exp H$ for some $H \in \mathfrak{s}$.*

Proof. By the torus theorem (S1), $f \in gSg^{-1}$ for some $g \in G$. As f is central, $f = g^{-1}fg \in S$. Since $\exp\colon \mathfrak{s} \to S$ is onto, $f = \exp H$ with $H \in \mathfrak{s}$. ∎

Lemma 8.2. *Assume the torus theorem (S1) and the covering theorem (S2). Then $\widetilde{T} = \exp_{\widetilde{K}} \mathfrak{t}$ is a maximal torus of $\widetilde{K}$. Every $f \in F$ equals $\exp_{\widetilde{K}} H$ for some $H \in \Lambda$.*

Proof. Since $dp = \mathrm{id}_{\mathfrak{k}}$, naturality of exp (Lecture 2) gives

$$p(\exp_{\widetilde{K}} X) = \exp_K X \qquad (X \in \mathfrak{k}). \tag{8.2}$$

Let T' be the identity component of the closed subgroup $p^{-1}(T)$. Its Lie algebra consists of the X with $\exp_K(tX) \in T$ for all t. This set is $\mathfrak{t}$. As $\mathfrak{t}$ is abelian, $\exp_{\widetilde{K}}|_{\mathfrak{t}}$ is a homomorphism. Its image is a subgroup of T' containing a neighbourhood of 1 (Lecture 2). So the image is open, hence also closed, in the connected group T'. Thus $\widetilde{T} = T'$.

So $\widetilde{T}$ is compact, connected and abelian, hence a torus. Let $S \supset \widetilde{T}$ be a torus. Then $\operatorname{Lie} S$ is abelian and contains $\mathfrak{t}$. Since $\mathfrak{t}$ is maximal abelian in $\mathfrak{k}$, $\operatorname{Lie} S = \mathfrak{t}$ and $S = \widetilde{T}$.

Finally, $f \in F$ is central in the compact connected group $\widetilde{K}$. By the central-element lemma, Lemma 8.1, $f = \exp_{\widetilde{K}} H$ with $H \in \mathfrak{t}$. By naturality (8.2), $\exp_K H = p(f) = 1$, so $H \in \Lambda$. ∎

Theorem 8.3. *Assume the structure inputs (S1)–(S4). Each irreducible representation of K has a unique highest weight. It lies in $X^*(T)^+$, and this gives a bijection $\operatorname{Irr}(K) \to X^*(T)^+$.*

Proof. Step 1: the highest weight. Let Π be irreducible on V, and let $Z \in \mathfrak{z}$. Since $[Z, X] = 0$ for every $X \in \mathfrak{k}$, the element $\exp tZ$ commutes with every $\exp X$. The elements $\exp X$ generate K by the torus theorem (S1), so $\exp tZ$ is central. Hence $\Pi(\exp tZ)$ commutes with $\Pi(K)$. Differentiating at $t = 0$, so does $d\Pi(Z)$. By Schur's lemma, $d\Pi(Z)$ is a scalar. The operator $d\Pi(Z)$ is skew-adjoint, so the scalar is imaginary. The scalar depends linearly on Z. Hence

$$d\Pi(Z) = i\mu(Z)I \qquad \text{for some } \mu \in \mathfrak{z}^*.$$

Let $W \subset V$ be a complex subspace invariant under $d\Pi(\mathfrak{k}_s)$. Then W is invariant under $d\Pi(\mathfrak{k})$, and so under every $\Pi(\exp X) = e^{d\Pi(X)}$. The elements $\exp X$ generate K by the torus theorem (S1), so W is K-invariant. Hence V is irreducible under $\mathfrak{k}_s$. The complexification $\mathfrak{k}_{s,\mathbb{C}}$ has the same invariant complex subspaces, so V is also irreducible under $\mathfrak{k}_{s,\mathbb{C}}$.

By the highest-weight theorem (S4), the $\mathfrak{k}_{s,\mathbb{C}}$-module V has a highest weight λ and a highest weight vector v. For $H = Z + H_s$ with $Z \in \mathfrak{z}$ and $H_s \in \mathfrak{t}_s$,

$$d\Pi(H)v = i\big(\mu(Z) + \lambda(H_s)\big)v = i\ell(H)v, \qquad \ell := \mu + \lambda \in \mathfrak{t}^*.$$

Exponentiating gives $\Pi(\exp H)v = e^{i\ell(H)}v$. So v is a common eigenvector of $\Pi(T)$, and $\ell \in X^*(T)$ by the description of characters in § 8.1.

Now let ℓ' be any T-weight of V. The restriction of ℓ' to $\mathfrak{z}$ equals μ. The restriction of ℓ' to $\mathfrak{t}_s$ lies in $\lambda - \mathbb{N}R^+$ by the highest-weight theorem (S4). Roots vanish on $\mathfrak{z}$, so $\ell - \ell' \in \mathbb{N}R^+$. Thus ℓ is the highest weight of Π. It is dominant, since $\langle \ell, \alpha^\vee \rangle = \langle \lambda, \alpha^\vee \rangle \geq 0$.

Step 2: injectivity. Let Π_1, Π_2 be irreducible with the same highest weight ℓ. They share μ and λ. By the uniqueness part of the highest-weight theorem (S4), there is an isomorphism $A\colon V_1 \to V_2$ of $\mathfrak{k}_{s,\mathbb{C}}$-modules. It also intertwines $d\Pi_j(Z) = i\mu(Z)I$. Hence for every $X \in \mathfrak{k}$

$$A\,\Pi_1(\exp X) = A\,e^{d\Pi_1(X)} = e^{d\Pi_2(X)}A = \Pi_2(\exp X)\,A.$$

The elements $\exp X$ generate K by the torus theorem (S1), so A is K-equivariant.

Step 3: existence. Let $\ell \in X^*(T)^+$, with $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. By the integrality computation in § 8.1, $\lambda \in P$, and λ is dominant. The module $V(\lambda)$ is only known to integrate to K_s. Since the map p has kernel F, we must show that F acts trivially. This step uses the hypothesis $\ell \in X^*(T)$, not only $\lambda \in P$.

The kernel $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$ of $\exp|_{\mathfrak{z}}$ lies in Λ. So $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, and

$$\chi_\mu(\exp Z) = e^{i\mu(Z)} \qquad (Z \in \mathfrak{z})$$

is a well-defined character of Z_0. By the highest-weight theorem (S4) and integration (S3), $V(\lambda)$ integrates to an irreducible representation Π_λ of K_s. Define on $V(\lambda)$

$$\widetilde{\Pi}(z,s) = \chi_\mu(z)\,\Pi_\lambda(s).$$

It is irreducible, because Π_λ already is. Let v be a highest weight vector. For $H = Z + H_s$ as above,

$$\widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\mu(Z)}e^{d\Pi_\lambda(H_s)}v = e^{i\mu(Z)}e^{i\lambda(H_s)}v = e^{i\ell(H)}v.$$

Let $f \in F$. By the lifting lemma, Lemma 8.2, $f = \exp_{\widetilde{K}} H$ with $H \in \Lambda$. As f is central, Schur's lemma gives $\widetilde{\Pi}(f) = cI$. Evaluating on v,

$$cv = \widetilde{\Pi}(\exp_{\widetilde{K}} H)v = e^{i\ell(H)}v = v,$$

because $\ell(\Lambda) \subset 2\pi\mathbb{Z}$. So F acts trivially. By the descent criterion (S3), $\widetilde{\Pi} = \Pi \circ p$ for a representation Π of K. It is irreducible, since p is onto.

It remains to identify the highest weight of Π. Since $dp = \mathrm{id}_{\mathfrak{k}}$, we have $d\Pi = d\widetilde{\Pi}$. So

$$d\Pi(Z) = i\mu(Z)I \quad (Z \in \mathfrak{z}), \qquad d\Pi|_{\mathfrak{k}_s} = d\Pi_\lambda.$$

Thus V is $V(\lambda)$ as an $\mathfrak{k}_{s,\mathbb{C}}$-module, with highest weight vector v. These are exactly the two data used in Step 1. Step 1 therefore gives the highest weight $\mu + \lambda = \ell$. ∎

Compactness gave the invariant inner product. This yields the weight decomposition of V. It also makes $d\Pi(Z)$ skew-adjoint, so μ is real. Compactness of $\widetilde{K}$ allowed the torus theorem (S1) inside the lifting lemma, Lemma 8.2. Schur's lemma needs only irreducibility. Connectedness entered through the torus theorem (S1). Connectedness is needed. The trivial character and det of $\mathrm{O}(2)$ have the same differential and differ on a reflection (Lecture 13).

H §§12.1–12.3 phrase the result with *analytically integral* elements. Hall's kernel lattice is our Λ divided by 2π. Hall obtains existence by analytic methods in Chapter 12. Our proof passes through the covering p.

8.3 Central characters and global forms

Let K_{sc} be compact, connected, simply connected and semisimple, with maximal torus T. Let F be a subgroup of its finite centre.

Corollary 8.4. *Assume the torus theorem (S1), integration and descent (S3), and the highest-weight theorem (S4). Let $\lambda \in P$ be dominant. Then $V(\lambda)$ descends to K_{sc}/F if and only if F acts trivially. Equivalently,*

$$\lambda(H) \in 2\pi\mathbb{Z} \qquad \text{whenever } H \in \mathfrak{t} \text{ and } \exp H \in F.$$

Proof. The first statement is the descent criterion (S3). Let Π be the representation of K_{sc} integrating $V(\lambda)$, which exists by integration (S3). Let $f \in F$. By the central-element lemma, Lemma 8.1, $f = \exp H$ with $H \in \mathfrak{t}$. By Schur's lemma, f acts by a scalar. Evaluating on a highest weight vector gives $\Pi(f) = e^{i\lambda(H)} I$. This is I exactly when $\lambda(H) \in 2\pi\mathbb{Z}$. ∎

Since $f \mapsto \Pi(f)$ is a homomorphism, it suffices to test generators of F.

Now allow a central torus. Here $F = \ker p$ again, as in the covering theorem (S2). Since F is central in $\widetilde{K}$, each $(z, s) \in F$ has s central in K_s.

We recall the notation of Step 3 in the proof of the classification theorem. The kernel of $\exp|_{\mathfrak{z}}$ is $\Lambda_{\mathfrak{z}} = \Lambda \cap \mathfrak{z}$. For $\mu \in \mathfrak{z}^*$ with $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, the character χ_μ of Z_0 is given by $\chi_\mu(\exp Z) = e^{i\mu(Z)}$. For dominant $\lambda \in P$, the representation of K_s integrating $V(\lambda)$ is Π_λ.

Let Π be an irreducible representation of a compact group. By Schur's lemma, each central element acts by a scalar. The map sending a central element to its scalar is a homomorphism from the centre to $\mathbb{C}^\times$, the *central character* of Π. Let c_λ be the central character of Π_λ. So $c_\lambda(s)$ is the scalar by which an element s of the centre of K_s acts on $V(\lambda)$.

Corollary 8.5. *Assume the structure inputs (S1)–(S4). Let $\mu \in \mathfrak{z}^*$ and $\lambda \in \mathfrak{t}_s^*$ satisfy*

(a) $\mu(\Lambda_{\mathfrak{z}}) \subset 2\pi\mathbb{Z}$, so χ_μ is a character of Z_0 (integral character);

(b) $\lambda \in P$ is dominant;

(c) $\chi_\mu(z)\, c_\lambda(s) = 1$ for all $(z, s) \in F$ (finite compatibility).

Then $\rho_{\mu,\lambda}(z, s) = \chi_\mu(z)\Pi_\lambda(s)$ descends to an irreducible representation of K. Every irreducible representation of K arises in this way from exactly one pair (μ, λ).

Proof. By the integrality condition (a), the dominance condition (b) and integration (S3), $\rho_{\mu,\lambda}$ is an irreducible representation of $\widetilde{K}$. Let $f = (z, s) \in F$. As noted before the corollary, s is central in K_s. So $\rho_{\mu,\lambda}(f) = \chi_\mu(z)c_\lambda(s)I$. By the descent criterion (S3), $\rho_{\mu,\lambda}$ descends if and only if the compatibility condition (c) holds. The descended representation is irreducible because p is onto. The computation in Step 3 of the proof of the classification theorem shows that the descended representation has highest weight $\mu + \lambda$.

Conversely, let Π be irreducible with highest weight ℓ. Put $\mu = \ell|_{\mathfrak{z}}$ and $\lambda = \ell|_{\mathfrak{t}_s}$. Step 3 proves the integrality condition (a) and the dominance condition (b). It also gives Π' with $\Pi' \circ p = \rho_{\mu,\lambda}$ and highest weight ℓ. By the injectivity in Step 2, $\Pi \cong \Pi'$. Since $\Pi' \circ p$ is trivial on F, the compatibility condition (c) holds. The pair is unique because $\mathfrak{t} = \mathfrak{z} \oplus \mathfrak{t}_s$ recovers (μ, λ) from $\mu + \lambda$. ∎

This corollary and the classification theorem, Theorem 8.3, show that the integrality, dominance and compatibility conditions together are equivalent to $\mu + \lambda \in X^*(T)^+$. The compatibility condition is the description of $X^*(T)$ in Lecture 7. Indeed, a central s lies in a maximal torus of K_s by the central-element lemma, Lemma 8.1. Evaluating on a highest weight vector shows that $c_\lambda(s)$ is the value at s of the character χ_λ. The U(2) example, Example 8.9, checks this directly.

A module of the reductive algebra $\mathfrak{k}_{\mathbb{C}}$ can fail to come from a representation of K, because compactness constrains the action of the centre.

Proposition 8.6. *Let Π be a representation of K and $Z \in \mathfrak{z}$. Then $d\Pi(Z)$ is diagonalisable with eigenvalues in $i\mathbb{R}$. If $\exp Z = 1$, these eigenvalues lie in $2\pi i\mathbb{Z}$.*

Proof. With respect to an invariant inner product, $\Pi(\exp tZ)$ is unitary, so $d\Pi(Z)$ is skew-adjoint. Hence it is diagonalisable with eigenvalues $i\theta_j$, $\theta_j \in \mathbb{R}$. If $\exp Z = 1$, then $e^{d\Pi(Z)} = \Pi(1) = I$, so $e^{i\theta_j} = 1$. ∎

For $\mathfrak{k} = \mathfrak{u}(1) = i\mathbb{R}$, neither of the following modules comes from U(1):

$$\rho_1(it) = t\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \rho_2(it) = \tfrac{i}{2}t.$$

The first is not semisimple. The second has the wrong period: $\exp(2\pi i) = 1$, but $\rho_2(2\pi i) = \pi i \notin 2\pi i\mathbb{Z}$. Both integrate to the noncompact simply connected group $\mathbb{R}$, where no averaging is available. In physics terms, a compact U(1) symmetry has integer charges, and a noncompact $\mathbb{R}$ allows any charge.

The same happens for $\mathfrak{gl}(2,\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}(2,\mathbb{C})$. Letting I act on V_m by any $s \in \mathbb{C}$ gives a module. It comes from U(2) if and only if $s \in \mathbb{Z}$ and $s \equiv m \pmod 2$ (Exercise 8.5). On the semisimple part, Weyl's complete reducibility theorem already gives semisimplicity and weights in P. The remaining constraint is $X^*(T)$, which can be smaller.

8.4 Global forms in examples

Each irreducible representation of K is one dot of the lattice $X^*(T)$ in the dominant region. Passing to a quotient of K removes dots. Figure 8.1 draws this for SU(2) and for SU(3). The labels (a, b) are defined in Example 8.8, and the marks are those of Lecture 7.

Example 8.7 (SU(2) and SO(3))**.** We continue the computation at the end of § 8.1, with $h = \operatorname{diag}(1, -1)$. Thus $\langle \ell, \alpha^\vee \rangle = \ell(ih)$, and $Q = \{\ell : \ell(ih) \in 2\mathbb{Z}\}$. The highest weight of V_m has $\ell(ih) = m$. Now $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm I\}$ (Lecture 4). The element $-I$ is central, and

$$\exp(\pi i h) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

By the quotient criterion, Corollary 8.4, $-I$ acts on V_m by

$$e^{i\ell(\pi i h)} = e^{i\pi m} = (-1)^m.$$

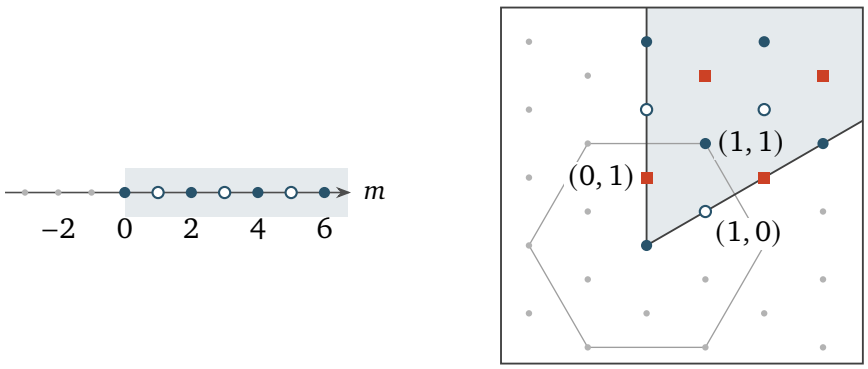


Figure 8.1: Characters of a maximal torus of SU(2), with coordinate $m = \ell(ih)$, on the left, and of SU(3) on the right, in the plane $\mathfrak{t}^*$. The shaded regions are the dominant regions, and the grey dots are not dominant. Filled dots in a shaded region are the highest weights of representations of SO(3) on the left and of PSU(3) on the right. On the right, a circle or a square marks the central value ω or ω^2. The thin hexagon joins the roots.

So V_m descends to SO(3) if and only if m is even. The irreducible representations of SO(3) have dimensions $1, 3, 5, \ldots$. In physics, V_m describes spin $m/2$. Hence the representations of integer spin descend to the rotation group SO(3). The representations of half-integer spin are representations of SU(2) that do not descend.

Let $r\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ be the covering. Its derivative is an isomorphism, and we identify the Lie algebras through it. Naturality of exp (Lecture 2) gives $\exp_{\mathrm{SO}(3)}(\theta\, ih) = r(\exp(\theta\, ih))$. This element is 1 if and only if $\exp(\theta\, ih) = \pm I$, that is, if and only if $\theta \in \pi\mathbb{Z}$. Hence the maximal torus $r(T)$ of SO(3) has $X^*(r(T)) = \{\ell : \pi\ell(ih) \in 2\pi\mathbb{Z}\} = Q$. This lattice has index 2 in P.

Example 8.8 (PSU(3))**.** Let $\mathfrak{t} = \{\mathrm{diag}(i\theta_1, i\theta_2, i\theta_3) : \sum \theta_j = 0\}$ and $\varepsilon_j(H) = \theta_j$. The roots are $\varepsilon_j - \varepsilon_k$ with $j \neq k$, with root vectors E_{jk}. We take the positive roots $\varepsilon_j - \varepsilon_k$ with $j < k$. The simple roots $\varepsilon_1 - \varepsilon_2$ and $\varepsilon_2 - \varepsilon_3$ have the real coroots $H_{12} = i\,\mathrm{diag}(1, -1, 0)$ and $H_{23} = i\,\mathrm{diag}(0, 1, -1)$. Then

$$\varepsilon_1(H_{12}) = 1, \quad \varepsilon_1(H_{23}) = 0, \quad (\varepsilon_1 + \varepsilon_2)(H_{12}) = 0, \quad (\varepsilon_1 + \varepsilon_2)(H_{23}) = 1.$$

The fundamental weights ϖ_1 and ϖ_2 are dual to the simple coroots. Thus ϖ_1 takes the values 1 and 0 at H_{12} and H_{23}, and ϖ_2 takes the values 0 and 1. So $\varpi_1 = \varepsilon_1$ and $\varpi_2 = \varepsilon_1 + \varepsilon_2$. The label (a, b) denotes the highest weight

$$\ell = a\varpi_1 + b\varpi_2 = (a + b)\varepsilon_1 + b\varepsilon_2.$$

The centre is $\{I, \omega I, \omega^2 I\}$ with $\omega = e^{2\pi i/3}$, and

$$\omega I = \exp H_\omega, \qquad H_\omega = \tfrac{2\pi i}{3}\,\mathrm{diag}(1, 1, -2) \in \mathfrak{t}.$$

Then

$$\ell(H_\omega) = (a + b)\tfrac{2\pi}{3} + b\tfrac{2\pi}{3} = \tfrac{2\pi}{3}(a + 2b),$$

so ωI acts on the irreducible module with highest weight (a, b) by ω^{a+2b}. Hence this module descends to $\mathrm{PSU}(3) = \mathrm{SU}(3)/\langle \omega I \rangle$ if and only if $3 \mid a + 2b$. Since $(a + 2b) - (a - b) = 3b$, this means $a \equiv b \pmod 3$. The dimension formula $\frac{1}{2}(a + 1)(b + 1)(a + b + 2)$ [I] gives the following values.

(a,b)	$(1,0)$	$(0,1)$	$(2,0)$	$(1,1)$	$(3,0)$	$(0,3)$	$(2,2)$
dimension	3	3	6	8	10	10	27
ωI acts by	ω	ω^2	ω^2	1	1	1	1

The adjoint representation $(1,1)$ descends, since $\mathrm{Ad}(\omega I) = \mathrm{id}$.

Example 8.9 (U(2)). Let $p\colon \mathrm{U}(1)\times\mathrm{SU}(2)\to\mathrm{U}(2)$ be given by $p(z,g)=zg$. The map p is onto. Indeed, given u, choose z with $z^2=\det u$ and put $g=z^{-1}u$. For the kernel,

$$zg=I \implies g=z^{-1}I, \quad 1=\det g=z^{-2}, \qquad \text{so} \qquad F=\{(1,I),(-1,-I)\}.$$

Here $\Lambda_{\mathfrak{z}}=2\pi i\mathbb{Z}\,I$, so the integral characters of $Z_0\cong\mathrm{U}(1)$ are $z\mapsto z^q$ with $q\in\mathbb{Z}$. Put $\Pi_{q,m}(z,g)=z^q\Pi_m(g)$. The compatibility condition (c) is

$$\Pi_{q,m}(-1,-I)=(-1)^q(-1)^m I=I \iff q+m \text{ is even.}$$

We cross-check with the classification theorem, Theorem 8.3. The torus $\mathrm{diag}(e^{i\theta_1},e^{i\theta_2})$ has $X^*(T)=\mathbb{Z}^2$, and (n_1,n_2) is dominant if and only if $n_1\ge n_2$. The torus element equals $p(e^{i\psi},\mathrm{diag}(e^{i\phi},e^{-i\phi}))$ when $\theta_1=\psi+\phi$ and $\theta_2=\psi-\phi$. On a highest weight vector of $\Pi_{q,m}$, this torus element acts by $e^{i(q\psi+m\phi)}$, while

$$n_1\theta_1+n_2\theta_2=(n_1+n_2)\psi+(n_1-n_2)\phi.$$

So $q=n_1+n_2$ and $m=n_1-n_2$. The numbers $n_{1,2}=(q\pm m)/2$ are integers exactly when $q+m$ is even. For example, $\mathbb{C}^2$ corresponds to $(q,m)=(1,1)$, and $\det$ corresponds to $(2,0)$.

8.5 Tensor products and consistency checks

Lemma 8.10. *Let f be central in K, acting by scalars c_1 on V_1 and c_2 on V_2. Then f acts by c_1c_2 on $V_1\otimes V_2$ and on each subrepresentation, and by c_1^{-1} on V_1^*.*

Proof. First, $(\Pi_1\otimes\Pi_2)(f)=c_1I\otimes c_2I=c_1c_2\,I$. Invariant subspaces inherit this scalar. On the dual, $\Pi_1^*(f)=\Pi_1(f^{-1})^T=c_1^{-1}I$. ∎

Hence the decompositions from [I] must respect central characters. We check this in the following cases.

- Clebsch–Gordan: $V_m\otimes V_n=\bigoplus_{k=0}^{\min(m,n)}V_{m+n-2k}$. Every summand has parity $m+n$, matching $(-1)^m(-1)^n$. In particular, $V_1\otimes V_1=V_2\oplus V_0$ is a sum of $\mathrm{SO}(3)$-representations, although V_1 is not an $\mathrm{SO}(3)$-representation.
- $\mathrm{SU}(3)$: $(1,0)\otimes(1,0)=(2,0)\oplus(0,1)$, with $9=6+3$ and $\omega\cdot\omega=\omega^2$ on both summands. Also $(1,0)\otimes(0,1)=(1,1)\oplus(0,0)$, with $9=8+1$ and $\omega\cdot\omega^2=1$.
- Duals: $V(a,b)^*\cong V(b,a)$ [I]. The lemma predicts $\omega^{b+2a}=\omega^{-(a+2b)}$. Indeed $(b+2a)+(a+2b)=3(a+b)$.
- Branching from $\mathrm{U}(2)$ to $\mathrm{SU}(2)$: $\Pi_{q,m}$ restricts to V_m. On the torus of $\mathrm{U}(2)$ it has the $m+1$ weights (n_1-j,n_2+j), $0\le j\le m$. Characters z^q multiply, so $\Pi_{1,1}\otimes\Pi_{1,1}=\Pi_{2,2}\oplus\Pi_{2,0}$. Both summands have $q+m$ even.

Self-checks

1. Does $V_3 \otimes V_5$ descend to $\mathrm{SO}(3)$? Decompose it.

   Answer. Yes. The element $-I$ acts by $(-1)^3(-1)^5 = 1$. By Clebsch–Gordan,

   $$V_3 \otimes V_5 = V_8 \oplus V_6 \oplus V_4 \oplus V_2, \qquad 9 + 7 + 5 + 3 = 24 = 4 \cdot 6.$$

2. Which irreducible representations of $\mathrm{PSU}(3)$ have dimension at most 10?

   Answer. The $\mathrm{SU}(3)$-irreducibles of dimension at most 10 are $(0,0)$, $(1,0)$, $(0,1)$, $(2,0)$, $(0,2)$, $(1,1)$, $(3,0)$, $(0,3)$. The next ones, such as $(2,1)$ and $(4,0)$, have dimension 15. The ones that descend to $\mathrm{PSU}(3)$ are those with $a \equiv b \pmod 3$, namely

   $$(0,0),\ (1,1),\ (3,0),\ (0,3),$$

   of dimensions $1, 8, 10, 10$.

3. Let $I \in \mathfrak{gl}(2,\mathbb{C})$ act by 1 on V_2. Does this module come from $\mathrm{U}(2)$?

   Answer. No. It would be $\Pi_{1,2}$, and $1 + 2$ is odd. Directly, compute $\Pi(-I)$ in two ways:

   $$-I = \exp(i\pi I) \implies \Pi(-I) = e^{i\pi \cdot 1} = -1, \qquad -I \in \mathrm{SU}(2) \implies \Pi(-I) = (-1)^2 = 1.$$

   These disagree.

Summary

Let K be a compact connected Lie group, and assume the cited structure inputs. An irreducible representation of K is determined by its highest weight. The possible highest weights are the dominant characters of a maximal torus T. For K_{sc}/F, test F on one highest weight vector. With a central torus, one adds an integral character of the central torus and a compatibility condition on F. In the examples, the descent conditions are m even for $\mathrm{SO}(3)$, $3 \mid a + 2b$ for $\mathrm{PSU}(3)$, and $q + m$ even for $\mathrm{U}(2)$.

Reading

H denotes Hall, *Lie Groups, Lie Algebras, and Representations*. BD denotes Bröcker–tom Dieck, *Representations of Compact Lie Groups*.

- H §§12.1–12.3: representations of compact groups, analytically integral elements, and the setting of the highest-weight classification for compact connected groups.
- H Chapter 13: fundamental groups of compact groups, and the comparison of algebraic and analytic integrality. This supports (S2).
- BD VI.2: the dominant weights of irreducible representations of compact connected Lie groups.
- Lie Algebras [I]: highest-weight theorem, Weyl's complete reducibility theorem, the dimension formula, Clebsch–Gordan, duals and $\mathfrak{sl}_3$ tensor products.